
PCS-9567MV-10000
Medium Voltage Skid
Installation Manual

NR Electric Co., Ltd.

Preface

About This Manual

The manual describes the PCS-9567MV-10000 Medium Voltage (MV) Skid in terms of safety information, product introduction, installation, and electrical connection, etc. Read all safety information and get familiar with the PCS-9567MV-10000 functions and features before proceeding with the operations of the PCS-9567MV-10000.

Intended Audience

This manual is intended for the persons who are required to install, configure or service equipment, or any other associated operation, it assumes a reasonable level of understanding in these disciplines.

Only qualified personnel are allowed to install, operate and maintain this equipment. Qualified personnel are individuals who:

- Are familiar with the installation, commissioning, and operation of the equipment and of the system to which it is being connected;
- Are able to safely perform switching operations in accordance with accepted safety engineering practices and are authorized to energize and de-energize equipment and to isolate, ground, and label it;
- Are trained in the care and use of safety apparatus in accordance with safety engineering practices;
- Are trained in emergency procedures (first aid).

Symbols

Safety information is highlighted with the following symbols according to the degree of danger:



Indicates an imminently hazardous situation that, if not avoided, will result in death or serious injury.



Indicates a potentially hazardous situation that, if not avoided, could result in death or serious injury.



Indicates a potentially hazardous situation that, if not avoided, may result in minor or moderate injury or equipment damage.

NOTICE

Indicates that property damage can result if the measures specified are not taken.

INFORMATION

Important information about the product, please pay attention to avoid undesired result.

Document Revision History

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The information in this manual is carefully checked periodically, and necessary corrections will be included in future editions. If nevertheless any errors are detected, suggestions for correction or improvement are greatly appreciated.

We reserve the rights to make technical improvements without notice.

NR ELECTRIC CO., LTD.

Headquarters: 69, Suyuan Avenue, Jiangning, Nanjing 211102, China

Manufactory: 18, Xinfeng Road, Jiangning, Nanjing 211111, China

Tel: +86-25-87178888

Fax: +86-25-87178999

Website: www.nrec.com/en

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1 Safety Instruction

This chapter presents the safety instructions for PCS-9567MV-10000 medium voltage skid (abbreviated as “MV skid”, or “this product”). Read this document thoroughly before operations. NR will not be liable for any damage caused by violation of the safety instructions in this document.

1.1 Personnel Requirements

- Only certified and professionally authorized personnel are allowed to work on or near the MV skid. All installation and service personnel must be trained to be familiar with the precautions and workflows mentioned in this manual, as well as the safety regulations.
- Strictly abide by all instructions contained in this manual, as well as the rules and laws of the country/region where the project is located.
- The minimum required safety equipment is as follows: safety shoes, safety gloves, safety glasses, head protection, work clothes. Take special precautions when you work close to exposed conductors.
- Only personnel with corresponding qualifications are permitted for special operations. Necessary safety precautions (such as fastening safety belts for climbing operations) must be taken.
- In case of abnormal conditions (such as abnormal noise, sudden bad weather, etc.), the operation should be stopped.
- Keep all doors locked while this product is operating. Spend as little time as possible near this product.

1.2 Mechanical Installation

- Only the transportation method described in the manual is allowed while transporting the MV skid. Please fully consider the weight of the MV skid and its gravity center.
- Please select certified and professionally authorized personnel to transport the MV skid.
- Please select the appropriate means of transport according to the site conditions, usually crane, and the selected means of transport must have sufficient load-bearing capacity.
- Please make sure that the strength of the sling used can meet the minimum required strength for lifting the MV skid.
- The crane should have sufficient arm length and rotation radius.
- Please make sure that all slings are secure and reliable, and the lengths of all slings connected to the corner fittings are the same.

1 Safety Instruction

- Please ensure that the harden road foundation of the freight passage has been completed. Transfer operations on soft or uneven foundations may cause damage to the MV skid and its accessories.
- Foundations for the MV skid should always be designed and constructed in accordance with local rules and laws, taking care that the foundations are properly planned and constructed. For example, improper foundations may cause MV skid settlement, or difficulty to open the doors.
- During the whole process of lifting the MV skid, it is necessary to operate in strict accordance with the safe operation procedures of the crane.
- It is forbidden to stand in the operating area, especially under the lifting arm or the lifting/moving machine, so as to avoid casualties.
- Please make sure that all doors of the MV skid are locked before transportation and movement
- The MV skid should be placed gently and stably after it is in place. It is forbidden to place the MV skid outside the vertical landing place by swinging the lifting appliance.
- Be sure to set warning signs or warning belts to prevent non-staff from entering the lifting and transportation area, so as to avoid accidents.
- Be careful NOT to drag on any surface when landing the MV skid. The ground should be firm and flat, well-drained and free of obstacles or protrusions.
- Do NOT throw the hooks onto the MV skid. This can damage the paint and cause corrosion or operation problems.

1.3 Electrical Connection

- Power cables and communication cables should be placed in different cable trenches. Communication cables should be laid as far away as possible from the power cables, and avoid power cables and communication cables walking parallel to each other for a long distance, so as to reduce electromagnetic interference caused by output voltage transients.
- There is large current in the power cable, so the damage of the cable insulation should be avoided during the installation process, so as to avoid short circuit. Moreover, the power cable should be fixed properly, and the connected cable should be carefully handled to avoid exerting excessive external force.
- Do NOT touch exposed terminals when the control device is energized, so as to avoid possible dangerous high voltage.
- Dangerous voltages may still be present in the DC circuit after the equipment power is turned off. These voltages will not disappear within several seconds.
- When connecting the AC voltage/current circuit or DC power supply circuit, please confirm the consistency with the nominal parameters of this product.

- When this control device is energized, it is forbidden to insert or unplug the PCB (Printed Circuit Board), otherwise it may cause the device to operate incorrectly.
- When connecting the output contacts of this control device to external circuit, the external power supply voltage should be carefully checked, so as to prevent the connected circuit from overheating.
- The grounding cable must be grounded reliably! Otherwise, this product may fail to work, and even may cause fatal shock hazard to the operator in case of malfunction.

1.4 Daily Operations

DANGER

The DC voltage may reach 1500V and the AC voltage may reach several tens of kilovolts when this product is running, and it will be dangerous to touch the live part of this product directly. Improper operation may lead to serious personal injury.

WARNING

Do NOT modify parameters of the MV skid without the permission of NR. Unauthorized modification of functional parameters may cause personal injury or equipment damage, in which case NR will not provide warranty service.

WARNING

When the primary system is in live operation, secondary side open circuit of the current transformer (CT) connected to this product is absolutely forbidden. Open circuit may cause extremely dangerous high voltages.

1.5 Maintenance & Replacement

- All maintenance and repair operations can only be performed on the equipment when the MV skid is completely disconnected from the battery system and the power grid, and it is confirmed that these power supplies will not be turned on again and the capacitor discharging is completed.
- Energy storage battery side open-circuit voltage must NOT exceed the maximum permissible input voltage, otherwise this product may be damaged.
- Grid-connected charging and discharging of MV skid should be approved by the local power supply department, and the relevant operation must be performed by certified and professionally authorized personnel.

1 Safety Instruction

- During the normal operation of the MV skid, do NOT disconnect the AC side circuit breaker (abbreviated as CB) or the DC side switch directly at will, otherwise personal injury or equipment damage may occur!
- Do NOT cut off the power supply of the PCS (Power Conversion System) control unit during operation. If it is required to cut off the power, please follow the shutdown operation process.
- Do NOT let the PCS run in the overload situation for a long time, please pay attention to the temperatures of the PCS and power modules in case of overload. If the temperature of the power module is high, please reduce the P (active power) value of the PCS remote regulation, so as to ensure the safety of operation.
- During maintenance, please make sure that the MV skid will not be energized accidentally. It is recommended to take insulation protection measures when contacting possible live parts.
- Clean the air filters in time to ensure smooth air flow.

1.6 Environmental Requirements

- The installation environment is outdoor.
- Please make sure that the installation environment is free from shaking and vibration.
- The foundation must meet the design requirements, ensuring it is solid, safe and reliable. The foundation must have the bearing capacity to support the weight of the MV skid. No depressions or inclinations are allowed on the ground surface. To prevent the risk of corrosion, the MV skid should be installed at a location higher than the surrounding environment, so that surface water does not accumulate around it.
- Adequate distance must be maintained between the installation location and walls or other equipment to meet the requirements for the maintenance access, escape route and ventilation.
- The installation environment should be as far away as possible from areas where people live, as the MV skid may generate some noise during operation.
- Ensure that all external cable entries are completely sealed to prevent the ingress of foreign objects such as rats, insects and dust.
- During storage and installation, avoid unnecessarily opening the doors to protect the equipment from rain and dust.

2 Product Introduction



Figure 2-1 Appearance of PCS-9567MV-10000 MV skid

Table 2-1 Technical Data of PCS-9567MV-10000 MV skid

Transformer	
Rated Power (kVA)	10000° / 9000° / 8000°
Transformer Model	Oil type ^[1]
Transformer Vector	Dy11-y11
Protection Level	IP54° / IP55°
Anti-corrosion Grade	C4-H° ^[2] / C4-VH° / C5-M° / C5-H° / C5-VH°
Cooling Method	ONAN° / ONAF°
Temperature Rise	60K (Top Oil) 65K (Winding) @40℃
Oil Retention Tank	None° / Galvanized steel° ^[3]
Winding Material	Aluminum° / Copper°
Transformer Oil	25# /45# mineral oil° / Natural ester insulation oil°
Transformer Efficiency	IEC standard° / IEC Tier-2°
MV Operating Voltage Range (kV)	11~33±5%
Nominal Frequency (Hz)	50 / 60
Altitude (m)	< 1000° / > 1000°
Switchgear	
Switchgear Type	Ring Main Unit, CCV ^[4]
Rated voltage (kV)	12/24/36
Insulating medium	SF6
Rated frequency (Hz)	50/60
Enclosure protection degree	IP3X
Gas tank protection degree	IP67
Gas leakage rate per year	≤0.1%
Rated Operating Current (A)	630

2 Product Introduction

Switchgear Short Circuit Rating (kA/s)	20kA/3s* / 25kA/3s°
Switchgear IAC (kA/s)	A FL 20kA 1S
PCS * 4	
DC Input Voltage Range (V)	1050~1500
Max. DC input Current (A)	1310*2
DC Voltage Ripple	< 2%
DC Current Ripple	< 3%
LV Nominal Operating Voltage (V)	690
LV Operating Voltage Range (V)	621~759
PCS Efficiency	98.7% ^[5]
Max. AC Output Current (A)	1151*2
Total Harmonic Distortion Rate	< 3%
Reactive Power Compensation	Four quadrant operation
Nominal Output Power (kVA)	1250*2
Max. AC Power (kVA)	1375*2
Power Factor Range	>0.99
Nominal Frequency (Hz)	50 / 60 Hz
Operating Frequency (Hz)	45~55 / 55~65 Hz
Connection Phases	Three-phase-three-wire
Protection	
DC Input Protection	Disconnecter + Fuse inside of inverter
AC Output Protection	Motorized Circuit breaker inside of Inverter
DC Overvoltage Protection	Surge arrester, type II* / I+II°
AC Overvoltage Protection	Surge arrester, type II* / I+II°
Ground Fault Protection	DC IMD* / DC IMD+ AC IMD°
Transformer Protection	Protection relay for pressure, temperature, gassing
Fire Extinguishing System	Smoke detector sensor (dry contact)
Communication Interface	
Communication Method	CAN / RS485 / RJ45 / Optical fiber
Supported Protocol	CAN / Modbus / IEC60870-103 / IEC61850
Ethernet Switch Qty	One for standard ^[6]
UPS	1kVA for 15min* / 1h° / 2h°
Skid General	
Dimensions (W*H*D)(mm)	12192*2896*2438 (40ft)
Weight (kg)	38800
Protection Level	IP54
Operating Temperature (°C)	-35~60, >45 derating
Storage Temperature (°C)	-40~70
Maximum Altitude (above sea level) (m)	5000, ≥3000 derating ^[7]

Environment Humidity	0~ 100%, No condensation
Type of Ventilation	Nature air cooling [•] / Forced air cooling [°]
Auxiliary Power Consumption (kVA)	21 (peak)
Auxiliary Transformer (kVA)	Without [•] / With [°] [8]

INFORMATION

● Standard ○ Optional

[1] If dry transformer is required, please contact with NR for more information.

[2] Lower protection level will be covered by C4-H.

[3] Standard for no supply of oil retention tank. If required to be integrated with PCS skid, please contact with NR.

[4] If other type of switchgear is required, please contact with NR for more information. In “**CCV**”, “**C**” is the abbreviation for “**Cable load break switch unit**”, “**V**” is the abbreviation for “**Vacuum circuit breaker unit**”.

[5] Typical discharge value for each PCS running at DC 1200V under IEC 62933-2-1 environment condition.

[6] If more Ethernet switch is required, please contact with NR for more information.

[7] When altitude is between 3000~4000m, the system LV AC voltage shall be less than 600V; When altitude is between 4000~4500m, the system LV AC voltage shall be less than 550V; When altitude is between 4500~6000m, please contact with NR for more information.

[8] Please contact with NR for more information.

3 Installation

3.1 Transport Requirements

- The transportation height should be as low as possible to ensure that the total transportation height does not exceed the maximum allowable height stipulated by local laws.
- Transport the equipment on the open heavy-duty chassis, not the enclosed trailer, for protecting the product surface from damage.

3.2 Arrival Inspection

- Check the appearance for damage. If any damage is found, please mark and record it, then open the equipment to check whether the interior is damaged. If there is any damage, please inform your local sales contact or on-site manager immediately.
- Repair any damaged paint.
- Check whether the equipment is in conformity with the delivery list and order. Record the deviation (if any) and immediately inform your local sales contact or on-site manager.
- Unloading
 - Protect the corners of the equipment from impact.
 - Verify the work surface is level and load-bearing compliant.
 - Remove all securing fixtures.
- Tools for unloading

Select an appropriate crane based on the product weight.

3.3 Storage Requirements

- The equipment should be placed on a solid and flat ground with adequate drainage, ensuring the area is clear of obstacles, vegetation or protrusions.
- Protect the equipment from dust and keep it away from humid, high temperature environments. During storage, ensure all cabinet doors remain securely locked, and avoid unnecessary door openings.
- Storage environment temperature: -40 °C~+55 °C; Relative humidity of storage environment: 0-95%, no condensation.
- Regular inspections, generally not less than once a week. Check if the packaging is intact and free from insect and rodent bites. If the outer packaging is damaged, it should be replaced

immediately.

- If on-site installation is not carried out immediately after completing the delivery acceptance work, the equipment must be stored according to the following requirements. Energy storage inverters with external packaging should be stored in a ventilated, dry, and clean indoor environment. Meanwhile, the following points should also be noted:

Before installing equipment that has been stored for a long time, the cabinet door should be opened for a thorough inspection to determine if the energy storage inverter is intact and undamaged. If necessary, professional personnel must conduct testing before installation.

The storage ground is flat and sturdy enough to support the weight of the energy storage inverter.

CAUTION

If condensation may occur in the storage area, please observe the following conditions.

- Power the internal heater to keep the internal temperature of the equipment higher than the external ambient temperature.
- If the power supply is unavailable on site, add desiccant bags inside the equipment.
- Pay attention to dealing with the harsh environment around, such as sudden cold, sudden heat, etc., which may cause equipment condensation, in order to avoid damage to the energy storage inverter.

3.4 Lifting Requirements

- Use specialized lifting equipment for hoisting.
- It is necessary to protect the corner of the MV skid from impact.
- It needs to be lifted through the four lifting points on the base of the MV skid.
- The sling shall not scratch the surface of the equipment, otherwise the damaged paint will lead to corrosion.
- Lift the equipment vertically, do not drag it on any surface.
- After the equipment is in place, put it down lightly and smoothly. It is forbidden to throw the equipment to places outside the vertical landing position.
- Be sure to set warning signs or warning belts to prevent non-staff from entering the lifting and transportation area to avoid accidents.
- When lifting, the tensile strength of wire rope must exceed 6 times the safety factor.
- When lifting, the equipment must be tested and can only be lifted in a completely safe state.

3 Installation

- When lifting, the equipment must be lifted smoothly.
- Ensure that all doors are locked before lifting and transportation.
- When lifting, please strictly follow the following diagram to use the lifting equipment for lifting operations.

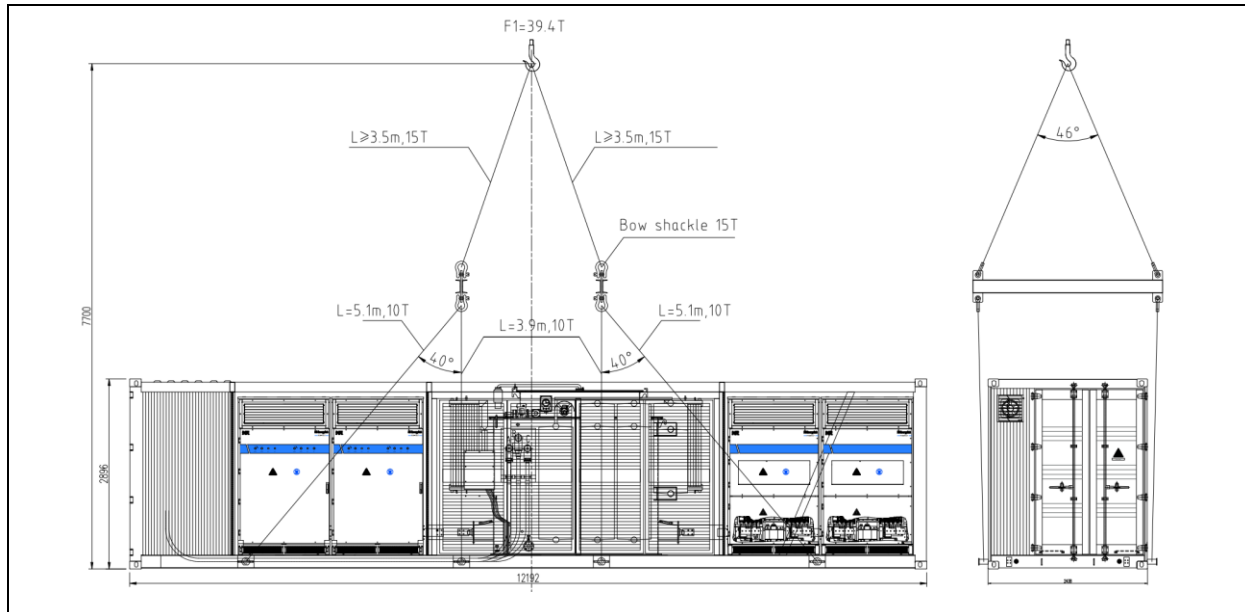


Figure 3-1 The diagram of lifting the MV skid

3.5 Construction Verification

3.5.1 Selecting Installation Location

- To prevent the risk of corrosion, install the MV skid at a location higher than the surrounding environment, so that surface water does not accumulate around it.
- Consider local conditions, such as soil type, frost protection and rainfall, etc., the foundation requires at least a certain amount of gravel.
- Consider the required cable bending radius and installation space.
- The whole periphery of the equipment must be located on the foundation.
- Check the bearing capacity of the ground and potential local special requirements (such as earthquake or typhoon anchoring). Use materials that are suitable for local conditions and requirements.

3.5.2 Foundation Requirements

- The hardened road foundation of the freight channel has been completed, otherwise the equipment and its accessories may be damaged during transportation on soft or uneven foundations.

- The cable trench has been laid.
- The pre-embedding of pipelines, cables, and grounding bars has been completed.
- The construction ground should have sufficient load-bearing capacity, and the construction space should meet the requirements for crane rotation.
- The civil structure, shape, dimensions and surrounding infrastructure comply with the blueprint and technical conditions.

3.6 Mechanical Installation

Before lifting the MV skid onto the foundation/platform, ensure that the foundation/platform is in good condition, hardened and stable.

- Ensure that the equipment is in direct contact with the foundation/platform smoothly and cannot be suspended.
- When the equipment is placed on the foundation, measure the height and inclination of the PCS, and check the slope of the ground surface around the PCS to ensure compliance with the engineering blueprint and technical specifications.
- Please follow local regulations and applicable standards to meet the mechanical connections and structural characteristics.
- The air outlets of equipment surrounding the PCS installation location should not blow towards the PCS. There should be no heat sources on the intake and exhaust sides of the PCS, and there should be no obstructions within a 1-meter radius.

NOTICE

For other dimensions and requirements not indicated, please refer to the engineering drawings provided.

4 Electrical Connection

4.1 Installation Tools

List of materials and tools required for electrical installation:

Equipment	PCS	
Tools	Wire stripper *1	
	Diagonal cutting pliers*1	
	Flathead screwdriver*1	
	Network Cable Crimper *1	
	Label printer *1	
	Heat-shrink tube printer *1	
	Adjustable wrench	
	20~100N·m torque wrench	
Materials	Network cable	See the site inventory for quantity
	RJ45 connector	Qty: As needed
	Copper cable lug	Qty: As needed
	Cable terminal	Qty: As needed
	Matching screw (For the DC primary cable)	Qty: As needed
	Heat-shrink marking sleeves	Qty: As needed
	Identification tags	Qty: As needed
	Sealing compound	Qty: As needed

4.2 Grounding Connection

According to the grounding instructions on the drawing, please use bolts to fix the grounding bar of the equipment to the grounding bar at the foundation with grounding markings.

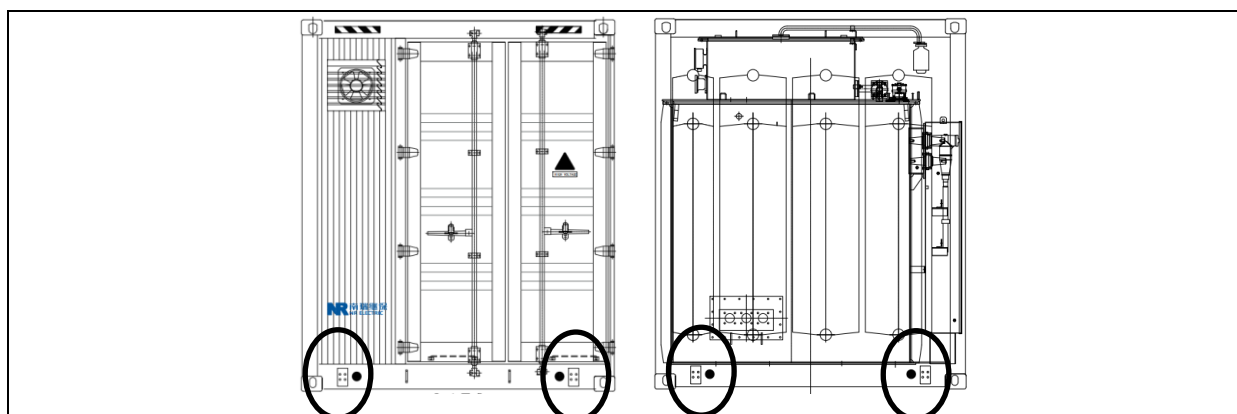


Figure 4-1 Grounding bar location diagram

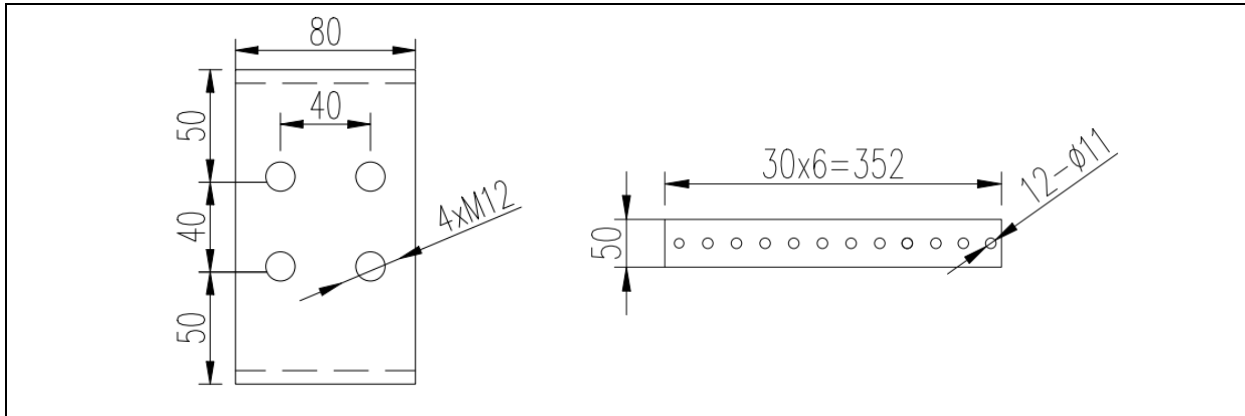


Figure 4-2 Parameters of MV skid cabinet grounding bar and bottom grounding bar

4.3 DC Side Connection

To make DC cable terminals, fasten them to the copper bar position at the output terminals of the PCS DC switch according to the diagram, taking care to distinguish between the positive and negative poles.

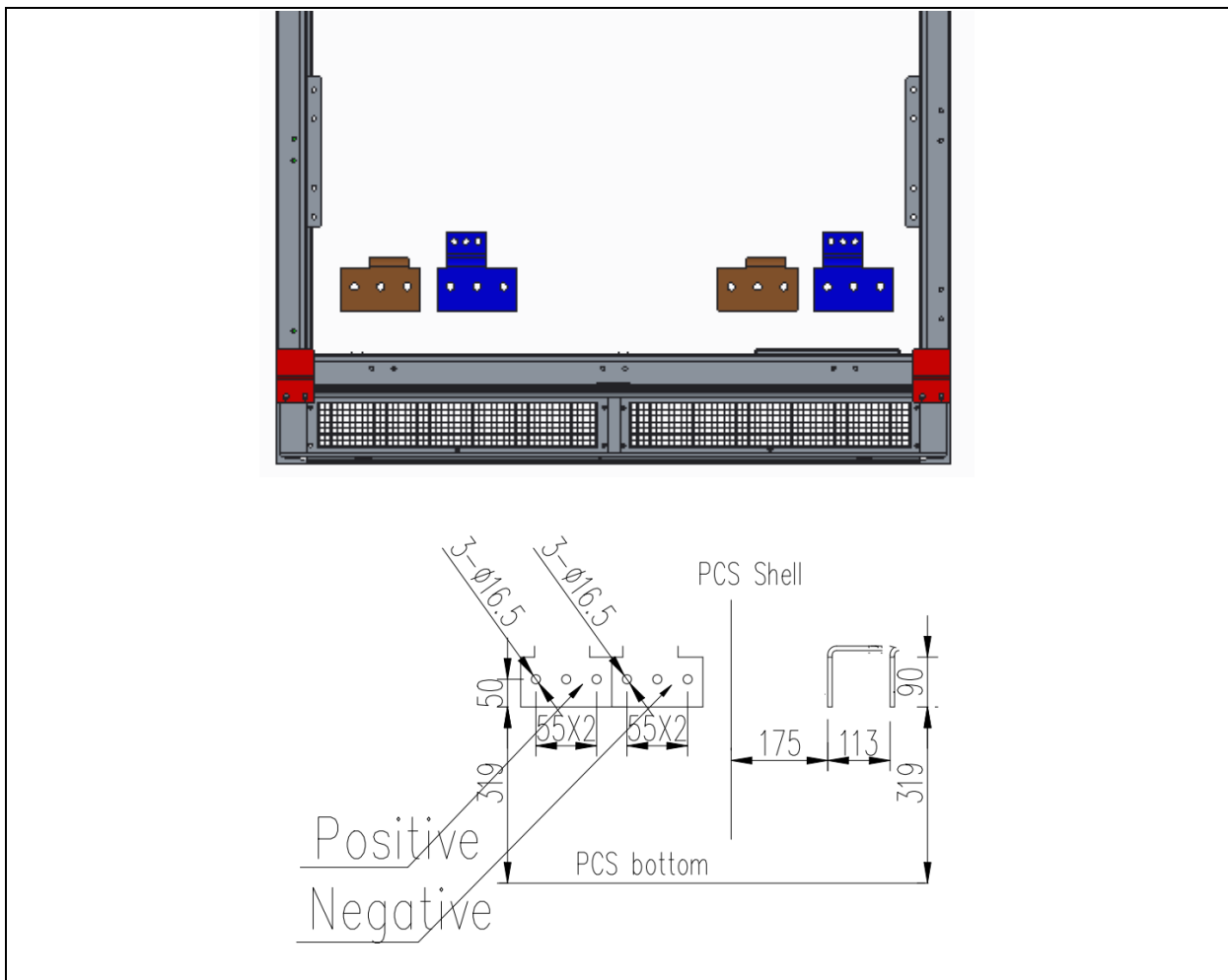


Figure 4-3 Diagram of DC side wiring position and parameters of cabinet

4.4 AC Side Connection

Make AC cable terminals, and secure them to the plug-in terminal ports of the AC incoming line on the RMU (Ring Main Unit) as shown in the diagram, taking care to distinguish the phase sequence.

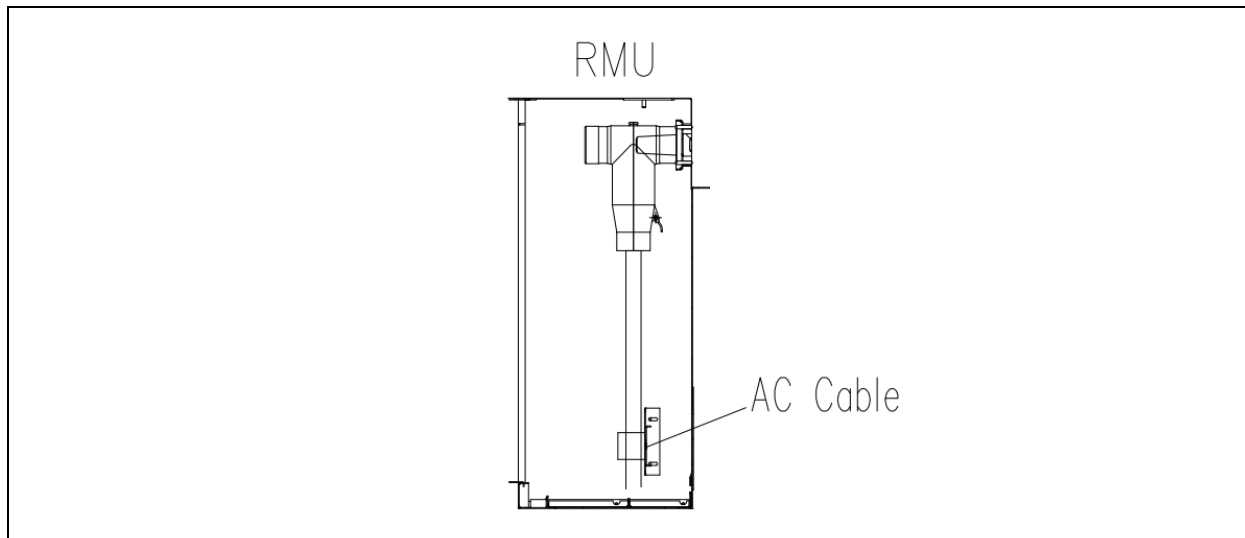


Figure 4-4 High voltage AC wiring diagram

⚠ CAUTION

The intrusion of moisture and sandstorms is highly likely to damage the equipment! To ensure the normal use of various functions of the equipment, please comply with the following:

Do not open the cabinet door when the air humidity is above 95%;

Avoid opening cabinet doors to install, maintain, or inspect equipment in rainy, damp, or windy weather conditions.

4.5 Communication Connection

4.5.1 PCS-to-EMS Communication

Make network cable connectors, connect one end of the network cable to the first and second network ports (Ports A & B) of the NR8191A plug-in module of the PCS control device, and connect the other end to the switch on station control layer.

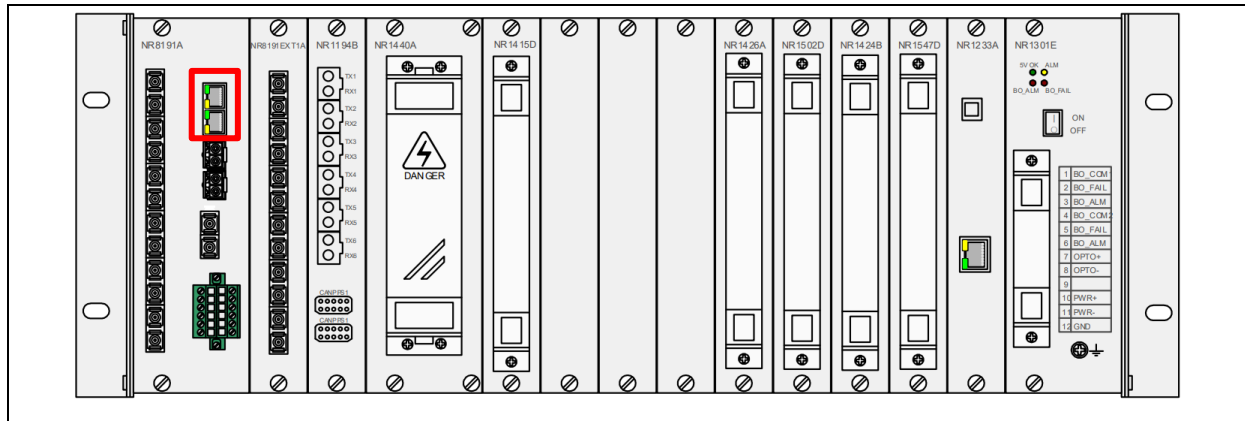


Figure 4-5 Communication interface diagram

4.5.2 PCS-to-BMS Communication

The PCS supports communication with the BMS via MODBUS TCP, MODBUS RTU, CAN, and other protocols. Connect to the corresponding positions indicated in the diagram below based on the communication protocol used.

MODBUS TCP wiring method: Connect the BMS communication network cable to the third network port of the NR8191A plug-in module of the PCS control device, as shown in the following figure:

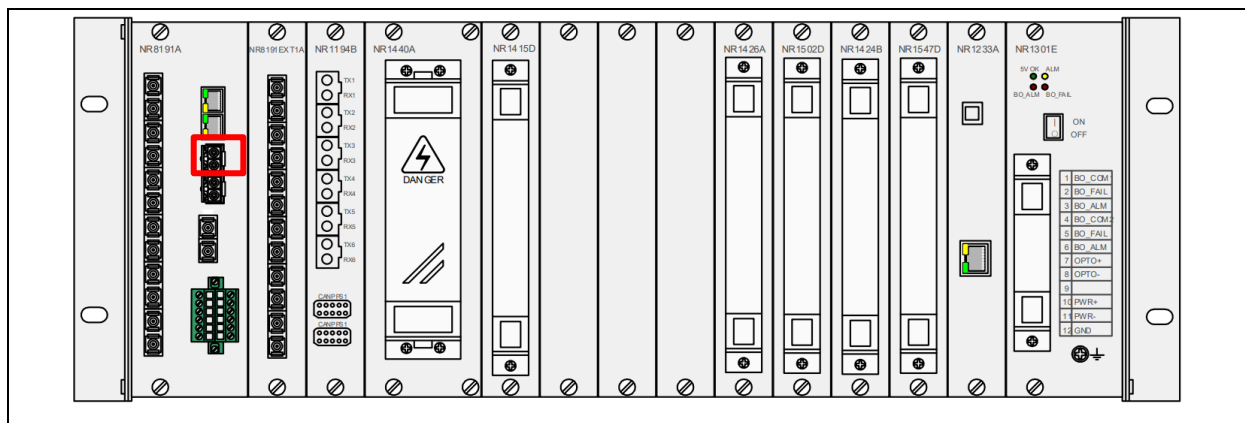


Figure 4-6 Communication interface diagram

MODBUS RTU wiring method: According to the drawing, connect the BMS communication serial cable to the corresponding terminal block of PCS.

CAN communication wiring method: Connect the BMS communication serial cable to ports 9, 10, and 11 of the NR8191A plug-in module's serial port terminals on the PCS device, as shown in the following figure:

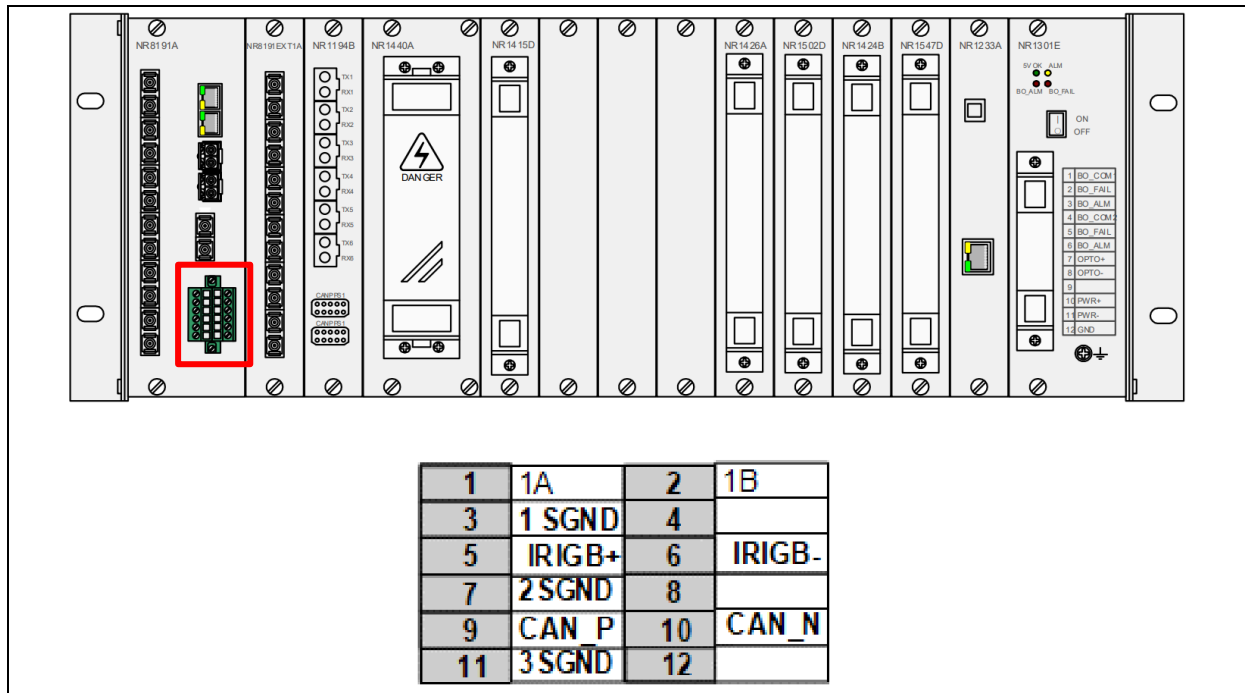
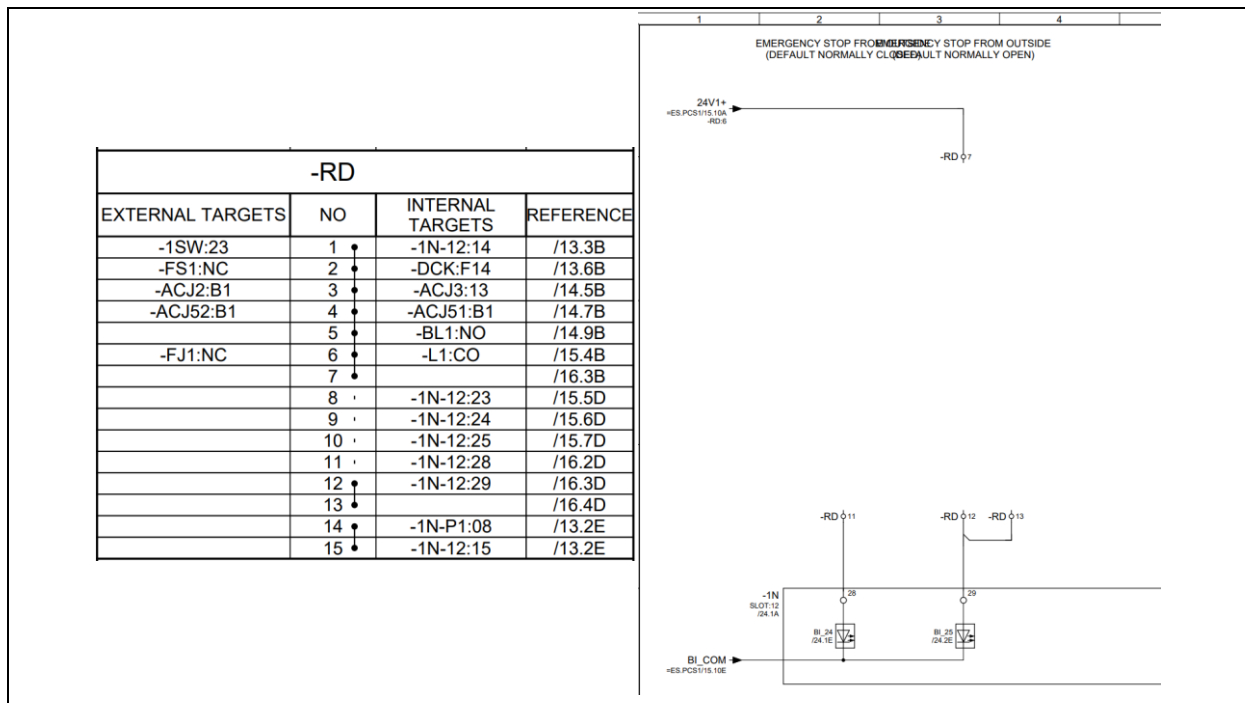


Figure 4-7 Communication interface diagram

BMS emergency stop hard contact wiring method: Connect the BMS emergency stop signal cable to the corresponding terminal block of PCS, and select normally open or normally closed terminals according to the actual node status on site, as shown in the following figure (refer to the actual project wiring drawing):



4.5.3 PCS-to-PMS Communication

Wiring between PCS and NR PMS: Connect the PMS communication fiber optic cable to the fourth port of the NR8191A module in the PCS device, as shown in the diagram below.

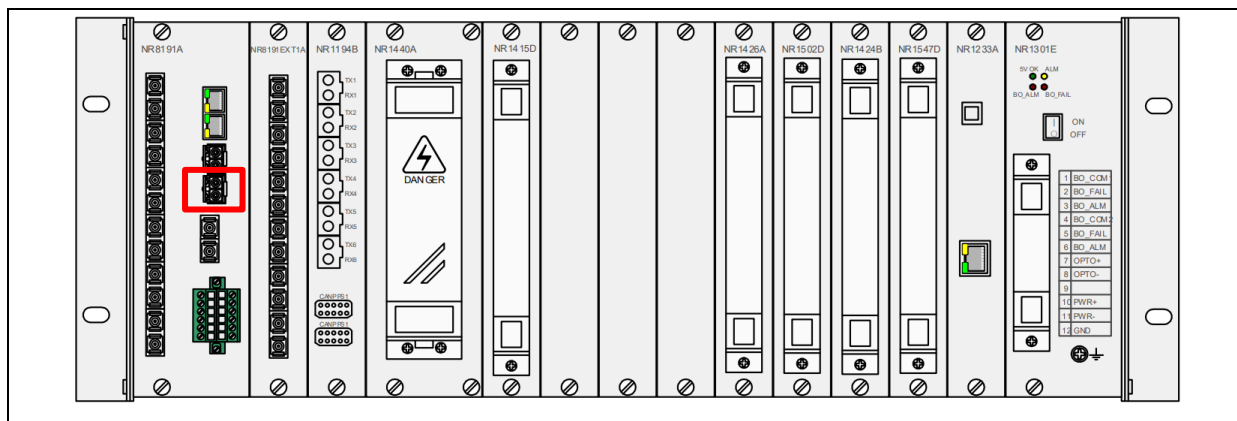


Figure 4-8 Communication interface diagram

4.5.4 PCS-to-PCS Communication

When two or more PCSs are connected to the same bus (AC or DC), carrier communication connection between PCSs is required. If two or more PCS are not on the same bus (AC or DC), carrier communication is unnecessary.

Select one PCS as the carrier master and the other PCS as slaves. The 9th, 10th, 11th, and 12th optical ports of the NR8191EXT1A plug-in module in slot 3 of the carrier master device are sequentially connected to the 8th optical port of the NR8191EXT1A plug-in module in slot 3 of the slave devices PCS1 to PCS4 through optical fibers.

4.5.5 PCS-9611S-to-EMS Communication

Connect the first and second network ports (Ports A & B) of the NR6106 plug-in module of PCS-9611S device to the EMS via network cable.

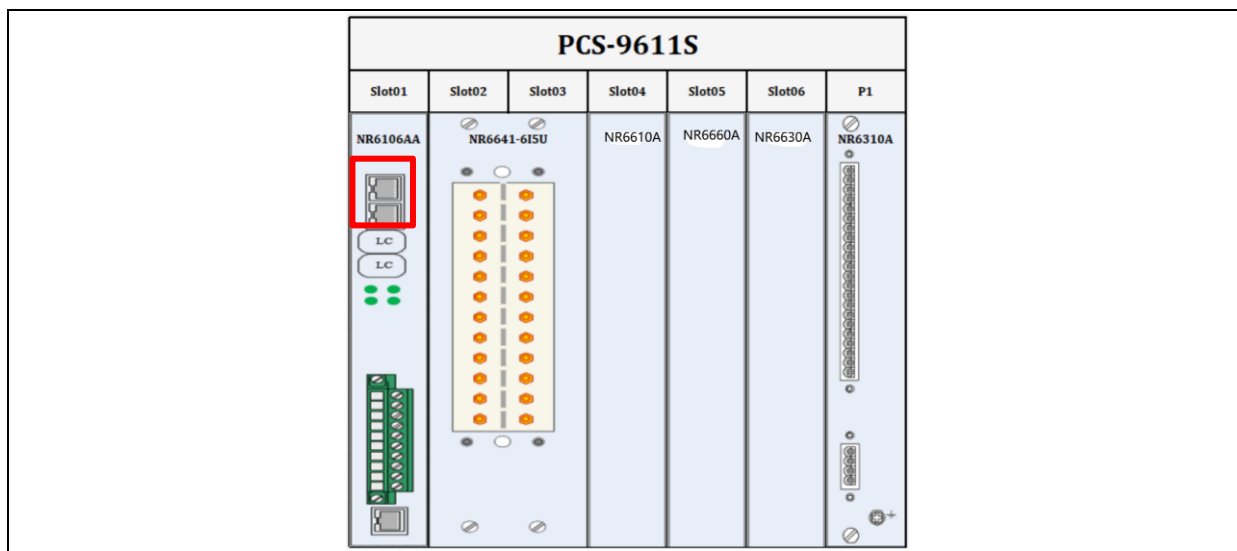


Figure 4-9 Communication interface diagram

4.6 Cable Labeling

Please follow the instructions below for cable labeling and electrical installation:

- To avoid stressing the copper bars of the outgoing cables from the PCS, a cable support structure should be considered outside the cabinet for the power cables entering and exiting the PCS.
- The onsite construction team should lay the DC primary cables and network cables according to the cable inventory, with a certain length of cable reserved.
- When laying the network cables, clearly mark the starting position of each cable. Our personnel will then perform the crimping work and connect the cables to the devices.
- The onsite construction team should prepare both ends of the DC primary cables according to the drawings and ensure that they are grounded at one end.
- The DC primary cables should be connected to the PCS DC copper bars using screws. Ensure tight contact between copper terminals and copper bars, securely fasten screws, and verify that no tension is applied to the cables after connection.
- Both network cables and DC primary cables should be labeled with tags.

4.7 Completion Inspection

As part of the pre-commissioning inspection of the equipment, it is necessary to conduct inspections after the mechanical and electrical installation are completed to ensure that the equipment is in good condition before commissioning.

To complete this inspection process, it is recommended to print the following inspection form and check it item by item. If any problem is found during inspection, user should contact the local sales contact or the site manager and handle it properly.

Inspection item	State	Remark
Mechanical installation		
Foundation construction		
Grounding and labeling		
Equipment fixed to foundation/platform		
No deformation or damage observed		
MV skid firmly fixed to foundation		
Electrical installation		
The primary AC cable connection is reliable, with correct phase sequence		
The DC cable connection is reliable, with correct polarity		
The connection ports of communication cables are correct and securely fixed		
The connection terminals of secondary cables are correct and securely fixed		
The cable labels are correct and clear		



The inlets have been sealed off		
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Appendix A Acronyms and Abbreviations

AC	Alternating Current
BCU	Bay Control Unit
BMS	Battery Management System
CB	Circuit Breaker
CT	Current Transformer
DC	Direct Current
EMI	Electromagnetic Interference
EMS	Energy Management System
GFM	Gird-Forming Mode
HV	High Voltage
IGBT	Insulated Gate Bipolar Transistor
LV	Low Voltage
MCB	Miniature Circuit Breaker
MV	Medium Voltage
PCB	Printed Circuit Board
PCS	Power Conversion System
RMU	Ring Main Unit
SOC	State of Charge
SOH	State of Health
SWG	Switchgear
THD	Total Harmonic Distortion
UPS	Uninterruptible Power Supply